



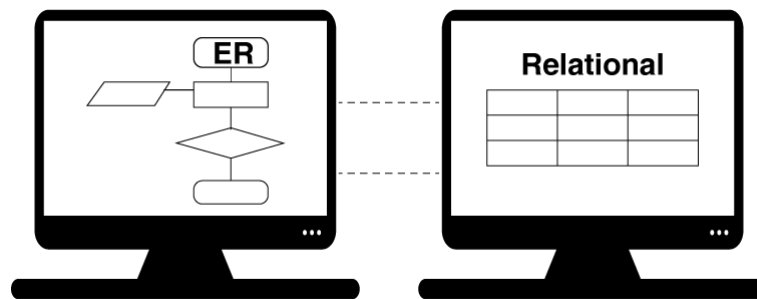
Instructor: Date:

Learner: Regional Branch:

Program: Center:

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**Let's Start!****Changing From One Model To Another****Learning Outcome:**

Exchange written and spoken information on principles, concepts, data base objects related to relational database management system in spoken or written texts.

Describe relational database, its concepts, and principles in spoken or written texts.

Ask your partner:

- In your opinion, what is the best way to organize data?
- Would you rather have data displayed on a map or on a table?

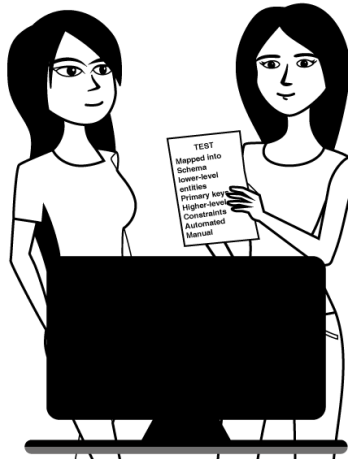
Can Do Statements:

- I can talk about turning from ER model to relational model.
- I can discuss how to turn diagrams from the ER model into tables in the relational model.
- I can talk about how to turn an ER hierarchical entity set into a table in the relational model



Let's Explore!

Sophie needs help with ER model and Relational model.



Instructions: Read the text below.

Macey: Hey Sophie! What's up?

Sophie: I'm worried about the test. Can you tell me how I can turn from the ER model to the relational model?

Macey: By all means! I have tons of information about that. Look, turning from the ER model to the relational model has to be done following two major steps. First, redundancy and irregularities have to be removed from the model you developed by making sure that the same data is not stored in multiple parts of the diagram. And second, the model has to be simplified to make it compatible to the relational model, so we need to convert or even remove certain attributes to fit in the relational model.

Sophie: Do you know if ER diagrams can be mapped into relational schema?

Macey: For sure they can! There are automated and manual ways to do it. First, entities in the ER diagram have to be converted into tables. So all entities that are represented by the rectangular box have to be changed to independent tables. Then, each attribute has to be converted into a column of the table.

Sophie: Cool! I have a specific question, could you tell me how I can turn an ER hierarchical entity set into a table in the relational model?

Macey: That one is pretty easy! First, you create a table for higher-level and lower-level entities. Then, primary keys of higher-level entities have to be added in the table of lower-level entities. After that, add all other attributes of lower-level entities in lower-level tables. Next, you declare primary key of higher-level table and the primary key for lower-level table. And finally, you declare foreign key constraints.

Sophie: I think I'm getting the hang of it now, thanks!

Macey: You're welcome!



Let's Practice!

Just two simple steps have to be followed to change from ER model to the relational model one.

Instructions: Read the paragraph. Write in the missing words. Use: have to be changed, has to be understood, has to be done, has to be, has to be performed. Read the paragraph aloud.

You may ask yourself what _____ to change the ER model to the relational one but you'll be pretty impressed on how fast you can get the hang of it. A process of two stages _____ followed to make such change happen. For starters, making sure repeated data is not stored in different parts of the diagram is the first step that _____ to be able to clean off redundancy and irregularities. The following step _____ pretty clear for you because it is the most important one; some attributes may _____, removed or simplified to make sure everything fits in the relational model.



Let's Practice!

Can you tell me how I can...

Instructions: Circle the correct answer. Practice reading the questions and sentences with your partner.

1. I have an ER hierarchical entity set. Would you _____ convert it into a table in the relational model?
 - a. tell me how can I
 - b. tell me how can
 - c. tell me how I can
 - d. tell me how could I
2. Can you let me _____ set the entities levels?
 - a. know I should
 - b. know how should
 - c. know how should I
 - d. know how I should
3. Could you specify _____ the level of the entities as higher-level and lower-level in the table?
 - a. where I have to set
 - b. where have to set
 - c. where have I to set
 - d. where I have set
4. _____ attributes of lower level entities have to be placed? Yes, in the lower-level tables.
 - a. Do you know where do the other
 - b. Do you know where the other
 - c. Do you know where does the other
 - d. Do you know where can the other
5. Can you identify _____
 - a. are primary key declared?
 - b. how primary key and key constrains are declared?
 - c. how are primary key and key constrains declared?
 - d. are primary key and key constrains declared?



**Extension Activity**

I can know more about ER Diagram and relational table.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look for more information about ER Diagram and relational table.
- Create a post with the main characteristics of ER Diagram and relational table.
- Share with your partners the post you created.

Self-Assessment**Yes****No**

- I look for more information about ER Diagram and relational table.
- I create a post with the main characteristics of ER Diagram and relational table.
- I share with my partners the post I created.

☐ ☐☐ ☐☐ ☐**Target Words**

- **for sure** (idiom): synonym for certainly or unquestionably
e.g. For sure you can! There are automated and manual ways to do it.
- **independent** (adjective): not connected with another or with each other; separate
e.g. entities that are represented by the rectangular box have to be changed to independent tables.
- **pretty** (adverb): synonym for quite or very
e.g. That one is pretty easy!
- **get the hang of something** (idiom): to learn how to do something
e.g. I think I'm getting the hang of it now, thanks!
- **tons** (noun): A large extent, amount, or quantity of something
e.g. I have tons of information about that.
- **by all means** (idiom): it has a similar meaning to that of absolutely
e.g. By all means! I have tons of information about that.



Learning Points

Passive of Modal Verbs with Have to

The **passive voice** is a form of speech in which you focus on the action itself rather than on the person who performs such action. **Passive voice** with modals is used to change the intention of the sentence: **have to** is a **modal verb** we use to express obligation, something that **has to be done**.

When we use the **passive voice** with **have to**, our main goal is to say that there is an **obligation** to do something we **need** to do.

To use the **passive voice** with **have to** you need to follow this structure:

Object + have/has to + verb be + main verb past participle + complement + subject (if known or necessary)

Example:

*These data **have to** be included in the report.*

We use **don't have to** to say that it is not necessary to do something, it is not an obligation. To use **don't have to** follow this structure:

Object + doesn't/don't have to + verb be + main verb past participle + complement + subject (if known or necessary)

Example:

*These data **don't have to** be modified by today, it's okay if you do it tomorrow.*

To form **questions** using **passive voice** and the modal verb **have to**, we use the auxiliary verb **do/does** in the **present simple**:

Do/does + object + have to + verb be + main verb past participle + complement + subject (if known or necessary)

Examples:

***Does** this software for compiling data **have to be** fixed?*

***Does** this software for compiling data **have to be** fixed?*

Examples

- *Redundancy and irregularities have to be removed from the model you developed by making sure that the same data is not stored in multiple parts of the diagram.*
- *Entities in the ER diagram have to be converted into tables.*
- *Each attribute has to be converted into a column of the table.*



Learning Points

Questions within questions

Questions within questions, or **indirect questions** are commonly used in polite situations or when talking to a person we don't know well.

These type of questions have two parts. The first part is usually an **introductory phrase**. The second part is the **indirect question**.

If you want to use **indirect questions**, there are some rules you need to remember:

To ask an **indirect question** using the verb **is** or **are** the verb comes after the subject. Not before the subject like in regular or normal questions.

Example:

Regular question :

Where **are** the data?

Indirect question :

Do you know where the data **are** ?

In **indirect questions** we don't use auxiliary verbs like **do, did** or **does** ; also the main verb is conjugated as if you were using it in an affirmative sentence.

Example:

Regular question :

What **does** he have to do to turn this ER model into a relational model?

Indirect question :

Can you tell him what he **has** to do to turn this ER model into a relational model?

Examples

- *Could you tell me how I can turn an ER hierarchical entity set into a table in the relational model?*
- *Do you know if ER diagrams can be mapped into relational schema?*